Phase A — Reproducing the High-Degree Sum-Product Construction

A source-disciplined reconstruction of $A = GP$

Central question

What property of one and the same set A makes both $A + A$ and AA small?

Independent study notes

26 August 2026

Phase A only. This document reproduces the known construction and contains no new candidate, improvement, or Phase B search.

How to read the source labels

Every substantive claim is marked by its status.

PAPER	A statement or estimate proved or stated in Bloom–Sawin–Schildkraut–Zhelezov. Lemma and theorem numbers refer to arXiv:2605.28781v1.
BACKGROUND	Standard algebraic number theory, geometry of numbers, or an elementary derivation used to unpack the paper.
INTERPRETATION	A conceptual synthesis, parameter reading, or reformulation. It is not presented as a separate theorem of the paper.
UNIT-DISTANCE ANALOGUE	A comparison with the supplied unit-distance proof, remarks, chain-of-thought record, or frozen guide.

The main mathematical source is Bloom et al. (2026). The unit-distance documents are comparison sources only; the rewritten chain-of-thought is not used as authority for any proof claim.

The letter c in the paper may denote different positive absolute constants at different occurrences. Here $c_0, C_0, C_1, \dots$ are disposable absolute constants, while D_0 denotes a fixed root-discriminant bound. No numerical value of the final power-saving constant is reproduced.

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1 Overview: the whole construction in one page

For a finite set $A \subset \mathbb{R}$, write

$$A + A = \{a + a' : a, a' \in A\}, \qquad AA = \{aa' : a, a' \in A\}.$$

The former measures additive expansion; the latter measures multiplicative expansion. The former sum-product conjecture over $\mathbb{R}$ asserted that every large finite set should have

$$\max\{|A + A|, |AA|\} \geq |A|^{2-o(1)}.$$

PAPER Bloom et al. (2026, Theorem 1.1) disproves this: there is an absolute $c > 0$ and arbitrarily large finite $A \subset \mathbb{R}$ such that

$$\max\{|A + A|, |AA|\} \leq |A|^{2-c}. \tag{1}$$

The construction takes a totally real field K of degree d, two lattice boxes

$$P = X + B^+(\varepsilon X) \subset \mathcal{O}_K, \qquad G = B^\times(Y) \subset \mathcal{O}_K^\times,$$

and then forms

$$\boxed{A = GP = \{gp : g \in G,\ p \in P\}.}$$

The set P is dense in a short box in ordinary Minkowski coordinates. The set G is dense in a box in logarithmic unit coordinates. The two extra compatibility facts are:

1. **Directness:** the map $G \times P \to GP$ is injective, so $|A| = |G||P|$.
2. **Bounded distortion:** every $g \in G$ obeys $e^{-Y} \leq |\sigma_i(g)| \leq e^Y$, so GP still lies in a controlled ordinary Minkowski box.

These facts yield the two decisive normalized estimates

$$\frac{|AA|}{|A|^2} \leq \frac{|GG|}{|G|^2} \leq c_0^{-d} Y^{1-d} \Delta_K^2, \tag{2}$$

$$\frac{|A + A|}{|A|} \leq (e^Y / c_0)^d \Delta_K^{1/2}. \tag{3}$$

PAPER These are the product and sum estimates in Lemma 4.1 of Bloom et al. (2026).

The central answer

INTERPRETATION After $\Delta_K \leq D_0^d$ and a sufficiently large fixed Y have been chosen, both sets are small because A is a *direct hybrid* of two dense boxes expressed in the coordinate systems suited to the two operations. The logarithmic box G supplies the factor $|GG|/|G|^2 \ll e^{-sd}$, for some fixed $s > 0$, in the product count. The ordinary box P, together with the coordinate-wise bound $|\sigma_i(g)| \leq e^Y$, keeps $A + A$ inside one box containing only $e^{O(d)} X^d$ algebraic integers. Neither component alone does both jobs.

2 The one-dimensional prototype and its obstruction

PAPER Section 2 of Bloom et al. (2026) begins with the Balog–Wooley prototype

$$A = GP,$$

where G is a short geometric progression and P is an interval. A geometric progression satisfies $|GG| \ll |G|$; an interval has small additive doubling.

The idea is attractive because

$$AA = (GG)(PP),$$

so multiplicative collisions inside G can reduce AA. Meanwhile, if all dilates gP remain in a short common interval, their sums can remain compressed.

INTERPRETATION The two separate facts $|GG| \ll |G|$ and $|P+P| \ll |P|$ do *not* by themselves control A. Three compatibility questions intervene:

1. Does $gp = g'p'$ occur often, making $|GP|$ much smaller than $|G||P|$?
2. How far do the dilates gP spread in the additive coordinate system?
3. Is the compression in GG large after normalization by $|A|^2$?

PAPER In one dimension the geometric progression is exponentially sparse. Making G longer increases its largest scale exponentially, so the common additive container for $GP+GP$ becomes too large. The resulting saving is only logarithmic, not $|A|^{-c}$ for a fixed $c > 0$ (Bloom et al., 2026, Section 2).

INTERPRETATION The replacement is not "a more complicated interval." It is a change in dimension. An additive box in $\mathbb{R}^d$ has lower-bound scale $X^d \Delta_K^{-1/2}$, and a unit-lattice box has lower-bound scale $Y^{d-1}/(\sqrt{d}R_K)$. A fixed gain or loss per coordinate is then raised to the power d. High degree makes a dense multiplicative set possible without requiring one enormous one-dimensional scale.

3 The minimal number-field toolkit

Let $K/\mathbb{Q}$ be totally real of degree d. Thus it has d embeddings $\sigma_1, \dots, \sigma_d : K \hookrightarrow \mathbb{R}$.

Object	Definition or standard fact	Why the proof needs it
Ring of integers $\mathcal{O}_K$	Algebraic integers in K; closed under addition and multiplication.	It is simultaneously an additive lattice and a ring, so the same finite set supports both operations.
Real embeddings	The d injective field maps $\sigma_i : K \to \mathbb{R}$.	They give coordinate-wise size control and eventually embed the construction in the target $\mathbb{R}$.
Minkowski embedding	$\Phi(\alpha) = (\sigma_1(\alpha), \dots, \sigma_d(\alpha))$.	It identifies $\mathcal{O}_K$ with a full-rank lattice in $\mathbb{R}^d$.
Discriminant Δ_K	The additive lattice $\Phi(\mathcal{O}_K)$ has covolume $\Delta_K^{1/2}$.	It is the density cost in counting algebraic integers in an ordinary box.
Unit group $\mathcal{O}_K^\times$	Algebraic integers whose inverses are also algebraic integers.	Units scale P without introducing denominators or leaving $\mathcal{O}_K$.
Logarithmic embedding	$\ell(u) = (\log\lvert\sigma_1(u)\rvert, \dots, \log\lvert\sigma_d(u)\rvert)$.	It converts multiplication into vector addition: $\ell(uv) = \ell(u) + \ell(v)$.

Object	Definition or standard fact	Why the proof needs it
Product-formula hyperplane	For a unit, $\prod_i \lvert\sigma_i(u)\rvert = 1$, hence $\sum_i \ell_i(u) = 0$.	The unit lattice has rank $d-1$ in $H = \{x : \sum x_i = 0\}$.
Regulator R_K	The logarithmic unit lattice modulo $\{\pm 1\}$ has covolume $\sqrt{d}R_K$ in H.	It is the density cost in counting units in a logarithmic box.

BACKGROUND These are standard facts from Minkowski theory and Dirichlet's unit theorem.
PAPER The paper records the additive covolume before Lemma 3.3 and the unit-lattice covolume before Lemma 3.5 (Bloom et al., 2026, Section 3).

Two uniform estimates make the toolkit quantitative.

PAPER First, for totally real K of degree $d \geq 2$, Lemma 3.1 proves

$$R_K \leq \Delta_K. \tag{4}$$

Thus a discriminant bound also controls the multiplicative lattice cost.

PAPER Second, Theorem 3.2, attributed there to Martinet, supplies an absolute $D_0 > 1$ and infinitely many degrees d with totally real K satisfying

$$\Delta_K \leq D_0^d. \tag{5}$$

This is exactly bounded root discriminant: $\Delta_K^{1/d} \leq D_0$.

4 The additive component P

4.1 The additive box and its lattice count

Define, as in the paper,

$$B^+(T) = \{\alpha \in \mathcal{O}_K : \lvert\sigma_i(\alpha)\rvert \leq T \text{ for } 1 \leq i \leq d\}. \tag{6}$$

Under the Minkowski embedding this is the intersection of $\Phi(\mathcal{O}_K)$ with the L^∞-box $[-T, T]^d$.

PAPER Lemma 3.3 gives, for $T \geq 1$,

$$T^d \Delta_K^{-1/2} \leq \lvert B^+(T)\rvert \leq (2T+1)^d. \tag{7}$$

BACKGROUND Here is the mechanism behind both sides.

- **Upper bound.** Distinct algebraic integers are 1-separated in the L^∞ norm. Indeed, if $x \neq y$, then $N_{K/\mathbb{Q}}(x-y)$ is a nonzero integer, so

$$1 \leq \lvert N_{K/\mathbb{Q}}(x-y)\rvert = \prod_{i=1}^{d} \lvert\sigma_i(x-y)\rvert.$$

  At least one factor is at least 1. Packing disjoint half-unit boxes gives at most $(2T+1)^d$ points.

- **Lower bound.** A box of side length T has volume T^d, whereas the lattice has covolume $\Delta_K^{1/2}$. Blichfeldt's lemma finds a translate of $[-T/2, T/2]^d$ containing at least $T^d/\Delta_K^{1/2}$ lattice points. Fixing one point and subtracting it from all the others gives the same number of distinct elements of $B^+(T)$.

The discriminant factor is therefore not decorative: it is exactly the additive lattice's covolume penalty.

4.2 The exact definition of P

Lemma 4.1 formally assumes real $X, Y \geq 2$. In its proof, after bounded ranges and rounding are absorbed into the absolute constants, the authors reduce without loss to sufficiently large X, Y and integral X. Work in that proof-normalized regime and fix a sufficiently small absolute $\varepsilon > 0$. The paper sets

$$\boxed{P = X + B^+(\varepsilon X).} \tag{8}$$

Here the rational integer X is embedded diagonally, so every $p = X + b \in P$ satisfies

$$X(1-\varepsilon) \leq \sigma_i(p) \leq X(1+\varepsilon) \qquad (1 \leq i \leq d). \tag{9}$$

In particular, every embedding of p is positive.

Translation preserves cardinality, so (7) gives

$$(\varepsilon X)^d \Delta_K^{-1/2} \leq |P| \leq (2\varepsilon X + 1)^d. \tag{10}$$

PAPER Equations (8)–(10) are the definition and count in the proof of Lemma 4.1 (Bloom et al., 2026).

What additive advantage does P provide?

INTERPRETATION The useful fact is not merely that $P+P$ is small. It is that $|P| \geq (\varepsilon X)^d \Delta_K^{-1/2}$ while P occupies a box of width $O(X)$ in every embedding. If multiplication by G expands each coordinate by at most e^Y, then all of $GP + GP$ still lies in one box of width $O(Xe^Y)$. Thus P is a dense additive carrier for every dilate used in A.

5 The multiplicative component G

5.1 The logarithmic unit box

For $Y \geq 1$, define

$$B^\times(Y) = \{u \in \mathcal{O}_K^\times : |\log|\sigma_i(u)|| \leq Y \text{ for } 1 \leq i \leq d\}. \tag{11}$$

The construction takes

$$\boxed{G = B^\times(Y).} \tag{12}$$

Because $\ell(uv) = \ell(u) + \ell(v)$,

$$GG \subseteq B^\times(2Y). \tag{13}$$

This is multiplicative box doubling converted into ordinary additive box doubling in H.

The same definition also gives the archimedean bounds

$$e^{-Y} \leq |\sigma_i(g)| \leq e^Y \qquad (g \in G,\ 1 \leq i \leq d), \tag{14}$$

and hence $G \subseteq B^+(e^Y)$.

PAPER These are the definitions and the "trivial, but crucial" additive boundedness observation preceding Lemma 3.4 of Bloom et al. (2026).

5.2 Counting G and GG

PAPER Lemma 3.5 states

$$Y^{d-1}d^{-1/2}R_K^{-1} \le |B^\times(Y)| \le 10(5Y+1)^{d-1}. \tag{15}$$

BACKGROUND The lower bound again comes from covolume. The region $H \cap [-Y/2, Y/2]^d$ has $(d-1)$-volume at least Y^{d-1}, while the unit lattice has covolume $\sqrt{d}R_K$. Blichfeldt gives many lattice points in a translate; taking differences corresponds to taking quotients of units and puts the results in $B^\times(Y)$.

For the upper bound one needs a field-independent separation scale. Let $\varphi = (1+\sqrt{5})/2$. Lemma 3.4 proves that if a unit u has $\varphi^{-1} < |\sigma_i(u)| < \varphi$ for every i, then $u = \pm 1$. Consequently, distinct logarithmic unit-lattice points are separated by at least $\log\varphi$ in L^∞, up to the two units u and $-u$ with the same logarithmic image. Packing yields the upper bound in (15).

Applying the upper estimate at $2Y$ and using (13) gives

$$|GG| \le 10(10Y+1)^{d-1} \le (C_0 Y)^{d-1}. \tag{16}$$

Combining (15), (16), and $R_K \le \Delta_K$,

$$\begin{aligned} \frac{|GG|}{|G|^2} &\le \frac{10(10Y+1)^{d-1}\, dR_K^2}{Y^{2d-2}} \\ &\le c_0^{-d} Y^{1-d} \Delta_K^2, \end{aligned} \tag{17}$$

where polynomial factors in d and fixed numerical constants are absorbed into c_0^{-d}.

What multiplicative advantage does G provide?

INTERPRETATION The normalized collision gain is $|GG|/|G|^2$. The numerator has one $(d-1)$-dimensional logarithmic box's worth of points, while $|G|^2$ has two. Ignoring covolume and constants, the ratio is $Y^{d-1}/Y^{2d-2} = Y^{1-d}$. The regulator contributes R_K^2, later paid through $R_K \le \Delta_K$.

6 Forming $A = GP$: directness and size

Define

$$\boxed{A = GP = \{gp : g \in G,\ p \in P\} \subset \mathcal{O}_K.} \tag{18}$$

The inclusion in $\mathcal{O}_K$ uses that g is a unit and p is an algebraic integer.

6.1 The collision audit

Suppose

$$g_1 p_1 = g_2 p_2.$$

Then

$$g_1/g_2 = p_2/p_1 \in \mathcal{O}_K^\times. \tag{19}$$

For every real embedding, (9) implies

$$\frac{1-\varepsilon}{1+\varepsilon} \le \sigma_i(p_2/p_1) \le \frac{1+\varepsilon}{1-\varepsilon}. \tag{20}$$

Choose ε small enough that

$$\frac{1+\varepsilon}{1-\varepsilon} < \varphi; \tag{21}$$

for example, any $0 < \varepsilon < (\varphi-1)/(\varphi+1)$ works. Lemma 3.4 then forces $g_1/g_2 = \pm 1$. The ratio p_2/p_1 is positive in every embedding, so it cannot be -1. Hence $g_1 = g_2$, and then $p_1 = p_2$.

Therefore the product map is injective and

$$\boxed{|A| = |G||P|.} \tag{22}$$

PAPER This is the directness argument in the proof of Lemma 4.1 (Bloom et al., 2026). **BACKGROUND** The explicit sufficient range in (21) is an elementary unpacking of "ε sufficiently small."

6.2 The size estimate

Multiplying the lower bounds for G and P,

$$\begin{aligned} |A| &\ge \frac{Y^{d-1}}{\sqrt{d}R_K}(\varepsilon X)^d \Delta_K^{-1/2} \\ &\ge \frac{(c_0XY)^d}{Y\Delta_K^{3/2}}, \end{aligned} \tag{23}$$

where $R_K \le \Delta_K$, the fixed ε, and $d^{-1/2} \ge c_1^d$ are absorbed into c_0^d. The upper bounds similarly give

$$|A| \le (XY/c_0)^d. \tag{24}$$

PAPER Equations (23)–(24) are the size bounds of Lemma 4.1.

INTERPRETATION Without (22), the lower bound would contain an unknown collision divisor M: $|A| \ge |G||P|/M$. The product estimate is normalized by $|A|^2$, so a loss $M = e^{\kappa d}$ would reappear as $e^{2\kappa d}$ and could erase the entire high-dimensional saving.

7 Why AA is small

Since multiplication in K is commutative,

$$AA = (GP)(GP) = (GG)(PP) \tag{25}$$

as sets. Therefore

$$|AA| \le |GG||PP| \le |GG||P|^2. \tag{26}$$

Using directness, $|A|^2 = |G|^2|P|^2$, so

$$\frac{|AA|}{|A|^2} \le \frac{|GG|}{|G|^2}. \tag{27}$$

Substituting (17),

$$\boxed{|AA| \le c_0^{-d} Y^{1-d} \Delta_K^2 |A|^2.} \tag{28}$$

PAPER This is exactly the product-set conclusion of Lemma 4.1 (Bloom et al., 2026).

Notice what was *not* required: the proof uses only the trivial estimate $|PP| \le |P|^2$. All nontrivial product compression comes from G.

INTERPRETATION The saving is the quotient $|GG|/|G|^2$. A generic set of $|G|$ scalars would have nearly $|G|^2$ products. The logarithmic box has only one box's worth of products, so its normalized ratio is approximately Y^{1-d}, before the regulator and constant costs. Directness transfers that same normalized saving to AA.

8 Why $A + A$ is small

Let $a = gp \in A$. From (9) and (14), for every embedding σ_i,

$$|\sigma_i(a)| = |\sigma_i(g)||\sigma_i(p)| \le e^Y (1+\varepsilon) X \le 2Xe^Y. \tag{29}$$

Thus $A \subseteq B^+(2Xe^Y)$, and the triangle inequality gives

$$A + A \subseteq B^+(4Xe^Y). \tag{30}$$

Applying the upper bound in Lemma 3.3,

$$|A + A| \le (8Xe^Y + 1)^d \le (C_0 Xe^Y)^d. \tag{31}$$

To normalize this by $|A|$, it is enough to use the P-part of A: since $1 \in G$, one has $P = 1 \cdot P \subseteq A$, and hence $|A| \ge |P|$. Therefore

$$|A| \ge (\varepsilon X)^d \Delta_K^{-1/2}.$$

Substitution in (31) yields

$$\boxed{|A + A| \le (e^Y/c_0)^d \Delta_K^{1/2} |A|.} \tag{32}$$

PAPER Equations (29)–(32) reproduce the sumset argument in Lemma 4.1 (Bloom et al., 2026).

INTERPRETATION The logarithmic box permits many units, but it forbids any one embedding from expanding by more than e^Y. Consequently the union of all dilates gP does not fragment into unrelated distant pieces: it remains in one ordinary box whose side length is larger by only the fixed factor e^Y in each coordinate. The cost is e^{Yd}, explicit in (32); it is later paid by choosing the additive scale X large enough.

9 Roles of the parameters and invariants

Symbol	What it controls	Benefit	Cost or constraint
X	Center and width scale of $P = X + B^+(\varepsilon X)$	Gives roughly X^d additive points; a large fixed X absorbs the C^d sumset cost relative to $\lvert A\rvert$	Lemma 4.1 formally assumes real $X, Y \geq 2$; its proof reduces to sufficiently large, integral X; X does not itself improve $\lvert AA\rvert/\lvert A\rvert^2$
Y	Radius of the logarithmic unit box	Gives roughly Y^{d-1}/R_K units and product ratio roughly Y^{1-d}	Causes coordinate distortion up to e^Y, hence additive cost roughly e^{Yd}
d	Degree and ambient dimension	Amplifies every fixed per-coordinate gain; makes $\lvert A\rvert$ exponential in d	Also raises all constants and discriminant costs to the d-scale
Δ_K	Covolume squared of the additive lattice	Small root discriminant keeps P dense	Costs $\Delta_K^{1/2}$ in additive counting and, after regulator substitution, Δ_K^2 in the normalized product bound
R_K	Covolume of the logarithmic unit lattice up to $\sqrt{d}$	Small regulator makes G dense	Appears quadratically in $\lvert GG\rvert/\lvert G\rvert^2$; the paper pays it through $R_K \leq \Delta_K$
ε	Relative thickness of the short positive window around X	Ensures ratios of points in P are uniformly near 1, proving directness	A fixed factor ε^d in $\lvert P\rvert$, absorbed into the exponential base c_0^d

INTERPRETATION The order of choice matters:

$$\boxed{\text{field bound } D_0 \longrightarrow \text{choose fixed } Y \longrightarrow \text{choose fixed large } X \longrightarrow d \to \infty.}$$

Choosing Y first buys product compression. Choosing X afterward pays the additive distortion caused by that Y. Increasing d then amplifies the resulting fixed balance.

10 Gain-versus-cost ledger

In this section all bounds are rigorous; $c_0, C_0 > 0$ are absolute constants that may change from line to line.

Quantity	Raw estimate	Structural reason	Normalized effect	Gain or cost
$\lvert P\rvert$	**PAPER** $(\varepsilon X)^d \Delta_K^{-1/2} \le \lvert P\rvert \le (2\varepsilon X+1)^d$	**INTERPRETATION** Dense ordinary Minkowski box (Lemma 3.3)	**INTERPRETATION** Supplies the X^d-scale inside $\lvert A\rvert$	**INTERPRETATION** Gain X^d; cost $\Delta_K^{1/2}$
$\lvert G\rvert$	**PAPER** $Y^{d-1}/(\sqrt{d}R_K) \le \lvert G\rvert \le 10(5Y+1)^{d-1}$	**INTERPRETATION** Dense box in the rank-$(d-1)$ logarithmic unit lattice (Lemma 3.5)	**INTERPRETATION** Supplies many multiplicative scalars without denominators	**INTERPRETATION** Gain Y^{d-1}; cost $R_K\sqrt{d}$
$\lvert A\rvert$	**PAPER** $\lvert A\rvert = \lvert G\rvert\lvert P\rvert$, hence $(c_0XY)^d/(Y\Delta_K^{3/2}) \le \lvert A\rvert \le (XY/c_0)^d$	**INTERPRETATION** Unit separation makes $G\times P\to GP$ injective (Lemmas 3.4 and 4.1)	**INTERPRETATION** No collision divisor is lost	**INTERPRETATION** Exact product of the two component sizes
$\lvert GG\rvert$	**PAPER** $\lvert GG\rvert \le 10(10Y+1)^{d-1}$	**INTERPRETATION** $GG \subseteq B^\times(2Y)$ is logarithmic box doubling	**PAPER** $\lvert GG\rvert/\lvert G\rvert^2 \le c_0^{-d}Y^{1-d}\Delta_K^2$	**INTERPRETATION** Main multiplicative gain Y^{1-d}; regulator/discriminant cost
$\lvert AA\rvert$	**PAPER** $\lvert AA\rvert \le \lvert GG\rvert\lvert P\rvert^2$	**INTERPRETATION** $AA = (GG)(PP)$ and $\lvert PP\rvert \le \lvert P\rvert^2$	**PAPER** $\lvert AA\rvert/\lvert A\rvert^2 \le \lvert GG\rvert/\lvert G\rvert^2$	**INTERPRETATION** Inherits all nontrivial saving from G
$\lvert A+A\rvert$	**PAPER** $\lvert A+A\rvert \le (8Xe^Y+1)^d$	**INTERPRETATION** $A+A \subseteq B^+(4Xe^Y)$ is the box-container mechanism	**PAPER** $\lvert A+A\rvert/\lvert A\rvert \le (e^Y/c_0)^d\Delta_K^{1/2}$	**INTERPRETATION** Near-linear in $\lvert A\rvert$; cost $e^{Yd}\Delta_K^{1/2}$

PAPER The displayed estimates and normalized inequalities are from Lemmas 3.1, 3.3–3.5, and 4.1 of Bloom et al. (2026). INTERPRETATION The structural-reason and gain-or-cost descriptions are conceptual readings of those estimates.

Under $\Delta_K \le D_0^d$, the two final rates become

$$\frac{|AA|}{|A|^2} \le Y\left(\frac{D_0^2}{c_0 Y}\right)^d, \tag{33}$$

$$\frac{|A+A|}{|A|} \le \left(\frac{e^Y D_0^{1/2}}{c_0}\right)^d. \tag{34}$$

The competition

INTERPRETATION **Gain:** choose Y so that the base $D_0^2/(c_0 Y)$ in (33) is less than 1. This creates an e^{-sd} saving in $AA/|A|^2$.
Cost: the same Y creates a fixed base $e^Y D_0^{1/2}/c_0 > 1$ in (34). Choose X sufficiently large so that $|A| \ge X^d$ makes this fixed C^d factor at most $|A|^\eta$.

11 Why increasing the degree changes the exponent

11.1 Theorem-level parameter choice

Fix a bounded-root-discriminant family $\Delta_K \le D_0^d$. The paper chooses a fixed Y sufficiently large in terms of D_0 and the absolute constant in Lemma 4.1; one displayed choice is $Y = 4D_0^2 c_0^{-1}$. For all sufficiently large d,

$$|AA| \le 2^{-d}|A|^2. \tag{35}$$

BACKGROUND To obtain the precise normalization in Theorem 1.2, distinguish its requested parameter $x \ge 1$ from the scale X_L used in Lemma 4.1. Choose a fixed $L \ge 1$, depending only on c_0, D_0, Y, and an integer X_L satisfying

$$Lx \le X_L \le (L+1)x.$$

Taking L sufficiently large and discarding finitely many eligible degrees gives

$$\frac{(c_0 X_L Y)^d}{Y D_0^{3d/2}} \ge x^d, \qquad |A| \le \left(\frac{X_L Y}{c_0}\right)^d \le (Cx)^d$$

for an absolute $C > 1$. The normalized sumset and product-set estimates are unchanged by this fixed rescaling, and rounding is absorbed into C. Renaming x back to X, Theorem 1.2 gives

$$X^d \le |A| \le (CX)^d, \qquad |A+A| \le C^d|A|, \qquad |AA| \le 2^{-d}|A|^2. \tag{36}$$

PAPER This is Theorem 1.2 of Bloom et al. (2026), deduced there from Lemma 4.1 and Theorem 3.2.

11.2 Turning e^{-sd} into a power of $|A|$

Fix an output parameter $0 < \eta < 1$, and take $X = C^{1/\eta}$. From $|A| \ge X^d$,

$$C^d = X^{\eta d} \le |A|^\eta,$$

so

$$|A + A| \le |A|^{1+\eta}. \tag{37}$$

From $|A| \le (CX)^d$,

$$\log|A| \le d(\log C + \log X) = d \log C(1 + \eta^{-1}) \le \frac{2d \log C}{\eta}.$$

Thus $d \ge \eta \log|A|/(2 \log C)$, and

$$2^{-d} \le |A|^{-c_1\eta}, \qquad c_1 = \frac{\log 2}{2 \log C} > 0. \tag{38}$$

Combining (35) and (38),

$$|AA| \le |A|^{2-c_1\eta}.$$

PAPER This is Corollary 1.3 of Bloom et al. (2026); the displayed logarithmic conversion is an elementary expansion of the paper's deduction.

Finally choose $\eta = 1/(1 + c_1)$. Then

$$1 + \eta = 2 - c_1\eta = 2 - \frac{c_1}{1 + c_1},$$

which gives (1) with an absolute positive saving.

INTERPRETATION If d were fixed and only $X \to \infty$, the factor 2^{-d} would be a constant, not $|A|^{-\delta}$. Growing degree is what makes $d \asymp \log|A|$, so a fixed per-degree saving becomes a power saving. This is the exact sense in which high dimension improves the balance.

12 Construction mechanism versus field supply

Construction mechanism, assuming K

PAPER Lemma 4.1 is field-local. Given one totally real K, its discriminant, and parameters X, Y, it constructs P, G, A and proves the size, sumset, and product-set bounds. Its inputs are:

- additive lattice count (Lemma 3.3);
- unit separation (Lemma 3.4);
- unit-lattice count (Lemma 3.5);
- regulator estimate $R_K \le \Delta_K$ (Lemma 3.1).

Number-field supply

PAPER Theorem 3.2 supplies infinitely many degrees with totally real fields satisfying $\Delta_K \le D_0^d$. Lemma 3.1 then automatically gives $R_K \le D_0^d$. No split prime, class-group fibre, or bounded-denominator construction is required for the final sum-product argument (Bloom et al., 2026, Section 5).

INTERPRETATION This separation is conceptually important. The combinatorial construction needs two lattices with controlled covolume. The arithmetic supply theorem guarantees that the covolumes grow only exponentially in the dimension, so their penalties can be compared with the exponential gains.

13 The precise unit-distance analogy

The comparison below is made only after the sum-product proof has been derived independently.

Unit-distance ingredient	Role there	Sum-product analogue	Role here	Quantitative effect
Totally real tower F_j, then CM fields $K_j = F_j(i)$, with growing f_j and bounded root discriminant	Supplies many embeddings while keeping class-number and lattice costs exponential in f_j	Totally real fields K with growing d, $\Delta_K \le D_0^d$, and $R_K \le \Delta_K$	Supplies both the additive and logarithmic lattices at controlled covolume	Converts fixed per-degree gain into a power of the final set size
Fixed rational primes splitting completely; sign ideals; largest class-group fibre	Produces $\|U\| \ge 2^{tf}/h(K) \ge e^{\gamma f}$ distinct directions	No corresponding split-prime step in the final proof; the full bounded-log unit box G is used directly	Produces $\|G\| \ge Y^{d-1}/(\sqrt{d}R_K)$ scalars	Unit distance: direction gain $e^{\gamma f}$; sum-product: product-collision gain $\|G\|^2/\|GG\|$
Norm-one family $U \subset Q^{-2}\mathcal{O}_K$	Each u is an allowed translation/difference, not the point set itself	Unit-lattice box $G \subset \mathcal{O}_K^\times$	Each g is a multiplicative scalar used in $A = GP$	In both cases a multiplicative family drives the extremal count, but its combinatorial role differs
Exact archimedean control $\|\sigma(u)\| = 1$	Translation by u has fixed length in every complex coordinate; product-disc overlap is ρ_R^f	Bounded logarithmic control $\|\log\|\sigma(g)\|\| \le Y$	Multiplication by g distorts each real coordinate by at most e^Y	Exact isometry becomes bounded distortion with explicit cost e^{Yd}
Minkowski lattice $Q^{-2}\mathcal{O}_K \subset \mathbb{C}^f$ and product-of-discs window	A torus average finds a finite coset window with many pairs $(x, x+u)$	Additive lattice $\mathcal{O}_K \subset \mathbb{R}^d$ and $P = X + B^+(\varepsilon X)$	Gives many points in a short positive box; all $GP+GP$ lies in a moderately enlarged box	Unit distance pays overlap ρ_R^f and packing; sum-product pays $\Delta_K^{1/2} e^{Yd}$
Class-group quotient and fixed denominator Q^2	Converts raw ideal choices into elements; the loss is $h(K)$, and Q^2 controls the lattice	No direct analogue: the final sum-product proof has a different cost ledger	Counts actual units, pays regulator/discriminant lattice-density costs, and separately proves $\|GP\| = \|G\|\|P\|$	Contrast: unit distance pays $h(K)$ and a fixed denominator; sum-product pays R_K, Δ_K costs and has no corresponding quotient
Projection to one complex coordinate	Injective on a lattice coset; turns every selected difference into a planar unit segment	Any real embedding $K \hookrightarrow \mathbb{R}$	Injective ring homomorphism; preserves sums, products, and cardinalities exactly	Neither injective map loses cardinality; unit distance separately loses at most a factor two when directed pairs become unordered edges
Gain $e^{\gamma f}$ versus point budget e^{Bf}	Gives $\nu(P) \ge \|P\|^{1+\delta}$	Product saving e^{-sd} versus size budget $\|A\| \le e^{Md}$	Gives $\|AA\| \le \|A\|^{2-s/M}$, while X absorbs the sumset cost	Same exponential-rate comparison, different extremal statistic

UNIT-DISTANCE ANALOGUE The unit-distance entries are sourced from Proposition 2.2, Theorem 2.3, and Lemmas 2.4–2.6 of OpenAI (2026), with the proof architecture cross-checked against Unit-distance guide (2026, Sections 6–12) and the human-digested overview in Alon et al. (2026).

UNIT-DISTANCE ANALOGUE INTERPRETATION Both proofs build in growing algebraic dimension and compare an exponential structural gain with exponential lattice costs. Unit distance obtains *exact norm-one translations* and counts surviving edges in a window; sum-product obtains *bounded-distortion unit scalars* and combines them directly with a dense additive box.

14 Reusable mechanisms extracted after reproduction

No new construction is proposed here. The purpose is only to classify what the reproduced proof actually uses.

Mechanism	What it accomplishes	Quantitative effect	Classification
Hybrid structure $A = GP$	Lets G carry product collisions while P carries additive density	$\lvert AA\rvert/\lvert A\rvert^2 \le \lvert GG\rvert/\lvert G\rvert^2$, while $A + A$ stays in one box	Essential; likely reusable
Different coordinates for different operations	Ordinary embeddings linearize addition; logarithmic embeddings linearize multiplication	Two independent lattice counts, with covolumes $\Delta_K^{1/2}$ and $\sqrt{d}R_K$	Essential; likely reusable
Bounded distortion	Permits multiplicative variation without uncontrolled additive spread	Coordinate cost at most e^Y, total box cost e^{Yd}	Essential here; likely reusable
High-dimensional amplification	Raises a fixed local gain or cost to the degree	If $\lvert A\rvert \le e^{Md}$, then $e^{-sd} \le \lvert A\rvert^{-s/M}$	Essential here; reusable when dimension can grow
Gain-versus-spread optimization	Forces every structural gain to be compared with its archimedean and covolume costs	Choose Y for product saving, then X to pay additive cost	Essential and broadly reusable
Construct algebraically, then embed in $\mathbb{R}$	Keeps both operations exact during construction and at the target	An injective real field embedding causes zero collision or operation loss	Essential for this target; reusable when a ring embedding exists

INTERPRETATION These classifications summarize the proof just reproduced. They do not assert that the mechanisms will succeed for a different extremal problem.

15 Can I reconstruct the proof without the paper?

The following is a proof-recall sheet, deliberately stripped of explanatory digressions.

Arrow	Mechanism in one sentence	Central inequality	Source
K	Choose infinitely many totally real fields of degree $d \to \infty$ with bounded root discriminant; control the regulator by the discriminant.	$\Delta_K \le D_0^d$, $R_K \le \Delta_K$.	Theorem 3.2; Lemma 3.1
$K \to P$	Count algebraic integers in an ordinary Minkowski box and translate it to a short positive window around X.	$P = X + B^+(\varepsilon X)$, $\lvert P\rvert \ge (\varepsilon X)^d \Delta_K^{-1/2}$.	Lemma 3.3; proof of Lemma 4.1
$K \to G$	Count units in a logarithmic box; multiplication becomes addition of log vectors.	$G = B^\times(Y)$, $\lvert G\rvert \ge Y^{d-1}/(\sqrt{d}R_K)$, $GG \subseteq B^\times(2Y)$.	Lemmas 3.4–3.5
$P, G \to A$	Ratios of two points of P are uniformly near 1; unit separation forces the product representation to be unique.	$A = GP$, $\lvert A\rvert = \lvert G\rvert\lvert P\rvert$.	Lemma 3.4; proof of Lemma 4.1
$A \to \lvert A\rvert$	Multiply the two lattice lower bounds and use $R_K \le \Delta_K$.	$(c_0XY)^d/(Y\Delta_K^{3/2}) \le \lvert A\rvert \le (XY/c_0)^d$.	Lemma 4.1
$A \to AA$	Factor products through GG and use directness to normalize by $\lvert G\rvert^2\lvert P\rvert^2$.	$\lvert AA\rvert \le c_0^{-d} Y^{1-d}\Delta_K^2 \lvert A\rvert^2$.	Lemma 4.1
$A \to A + A$	Bounded log coordinates imply $A + A$ lies in one ordinary box of radius $4Xe^Y$.	$\lvert A + A\rvert \le (e^Y/c_0)^d \Delta_K^{1/2}\lvert A\rvert$.	Lemma 4.1
Parameters	Fix Y large enough to make the product factor decay, fix X large enough to absorb the additive factor, then send $d \to \infty$.	$X^d \le \lvert A\rvert \le (CX)^d$, $\lvert A + A\rvert \le C^d\lvert A\rvert$, $\lvert AA\rvert \le 2^{-d}\lvert A\rvert^2$.	Theorem 1.2
Embedding	Apply any real embedding of K; it is an injective ring homomorphism.	All set cardinalities and both operations are preserved.	Theorem 1.2 to Corollary 1.3
Optimization	Take $X = C^{1/\eta}$; convert C^d and 2^{-d} into powers of $\lvert A\rvert$, then balance exponents.	$\lvert A+A\rvert \le \lvert A\rvert^{1+\eta}$, $\lvert AA\rvert \le \lvert A\rvert^{2-c_1\eta}$.	Corollary 1.3; Theorem 1.1

Minimal memory chain

INTERPRETATION

$$K \longrightarrow (B^+, B^\times) \longrightarrow (P, G) \longrightarrow A = GP,$$
$$A \longrightarrow \lvert A\rvert \longrightarrow (\lvert A + A\rvert, \lvert AA\rvert) \longrightarrow (Y, X, d) \text{ optimization.}$$

Remember the two indispensable bridges: *unit separation gives directness*, and *bounded logarithmic coordinates give additive bounded distortion.*

16 Ten-question self-test

Do not consult the preceding sections until you have attempted all ten questions.

1. Starting from the Minkowski embedding of $\mathcal{O}_K$, derive both sides of $T^d\Delta_K^{-1/2} \le |B^+(T)| \le (2T+1)^d$. Why is the lower bound obtained from a translated half-size box and a difference set?

2. Why is P defined as $X + B^+(\varepsilon X)$ rather than simply $B^+(X)$? Identify the exact later step that uses positivity and the short ratio window.

3. Why does putting units in a logarithmic L^∞-box make GG small? State both the containment for GG and the lattice-covolume estimate for $|G|$.

4. Reproduce the proof that $g_1p_1 = g_2p_2$ forces $g_1 = g_2$ and $p_1 = p_2$. Where are Lemma 3.4, total reality, and the smallness of ε used?

5. Starting from $AA = (GG)(PP)$, derive $|AA|/|A|^2 \le |GG|/|G|^2$. What would go wrong if GP were not direct?

6. Explain why the product saving is approximately Y^{1-d}, and list every cost that changes this heuristic into the rigorous factor $c_0^{-d}Y^{1-d}\Delta_K^2$.

7. Derive the coordinate bound $|\sigma_i(gp)| \le 2Xe^Y$, the container for $A+A$, and the final relative estimate for $|A+A|/|A|$.

8. Why must Y be chosen before X? Describe precisely which inequality Y improves and which inequality it worsens.

9. Why does fixed degree fail to turn 2^{-d} into a power saving in $|A|$? Using $X^d \le |A| \le (CX)^d$, derive the conversion from 2^{-d} to $|A|^{-c\eta}$ when $X = C^{1/\eta}$.

10. Give the closest unit-distance analogue of each of the following: P, G, bounded distortion, the discriminant/regulator cost, direct projection to $\mathbb{R}$, and the exponential-in-degree gain. For each, state one important difference as well as the similarity.

You are ready to leave Phase A only if you can reconstruct, without looking up the proof, the definitions of P, G, A, the directness argument, equations (28) and (32), and the conversion in (37)–(38). No new construction search is part of this document.

References

References

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Unit-distance rewritten chain of thought. Supplied discovery-stage record. Used only to understand attempted routes and not as authority for proof statements.